Implementing Gradient Descent

From slope to the next point

Hand-calculation worksheet

Name _____ Date _____

01 One Update

Shared Data and Model

Input x	-1	1
Target y	-2	2

$f_w(x) = wx$ $m = 2$ $b = 0$

Fix $b = 0$ and change only w One cross-section of the 3D cost surface

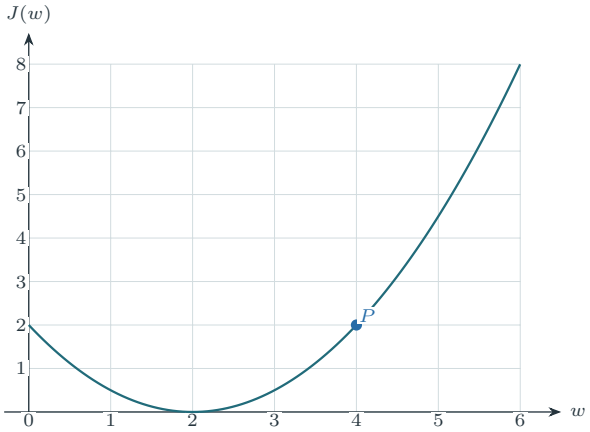
1 Calculate the predictions and cost at $w = 4$

$$J(w) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_w(x^{(i)}) - y^{(i)}\right)^2 = \frac{(-w+2)^2 + (w-2)^2}{4} = \frac{1}{2}(w-2)^2$$

$f_4(-1) =$ _____, $f_4(1) =$ _____

$$J(4) = \frac{(-4+2)^2 + (4-2)^2}{4} = \text{_____}$$

Blue point P is $(w, J(w)) = (4, 2)$



2 Use the sign of the slope to choose the direction of the update

$$J'(w) = \frac{d}{dw}J(w) = w - 2 \quad (\alpha > 0)$$

Current w	Slope $w-2$	Direction of the next w
0	_____	☐ Decrease ☐ Increase ☐ Unchanged
2	_____	☐ Decrease ☐ Increase ☐ Unchanged
4	_____	☐ Decrease ☐ Increase ☐ Unchanged

3 Starting at $w = 4$ with $\alpha = 0.5$, make one update and mark the new point Q above

$w_{k+1} = w_k - \alpha J'(w_k)$ current value $\longrightarrow$ next value

* w_0 is the starting value; w_1 is the first updated value Use the given derivative

$w_{\text{new}} = 4 - 0.5 \times$ _____ $=$ _____ $J(w_{\text{new}}) =$ _____

Reference · Supplied Gradient Descent algorithm and Intuition slides · Update rule and slope signs through C1 W1 slide 65

02 Learning Rate A

$J(w) = \frac{1}{2}(w-2)^2$ $J'(w) = w - 2$ $w_0 = 6$ $\boxed{\alpha = 3}$

4 Use the same rule to make **five updates**

$w_{k+1} = w_k - 3(w_k - 2)$

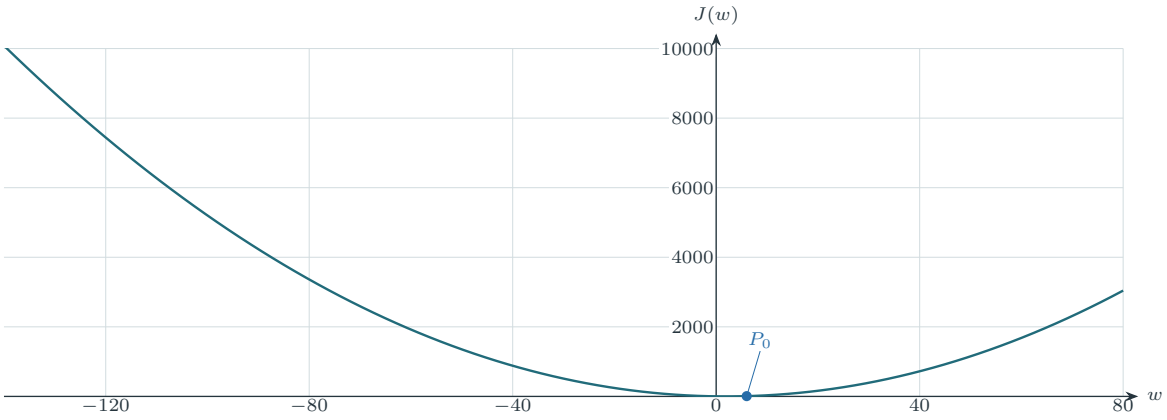
* Use the last value in a row as the current w in the next row

k	Current w_k	Slope $w_k - 2$	Subtract $\alpha(w_k - 2)$	Next w_{k+1}
0	6			
1				
2				
3				
4				

5 Calculate each cost, plot $P_k = (w_k, J(w_k))$, and draw arrows in order $P_0 \to P_1 \to \cdots \to P_5$

* $J(w_k) = \frac{1}{2}(w_k - 2)^2$ $P_0 = (6, 8)$

$J(w_1)$	$J(w_2)$	$J(w_3)$	$J(w_4)$	$J(w_5)$



6 Check the **position and cost** after five updates

$w_5 =$ _____ $|w_5 - 2| =$ _____

$J(w_5) = \frac{1}{2}(w_5 - 2)^2 =$ _____ $J(w_0) = 8$

☐ Closer ☐ Farther away ☐ Almost no movement

Explain why the cost increased even though you moved opposite to the slope

03 Learning Rate B

$J(w) = \frac{1}{2}(w-2)^2$ $J'(w) = w - 2$ $w_0 = 6$ $\boxed{\alpha = 0.01}$

7 Use the same rule to make **five updates**

$$w_{k+1} = w_k - 0.01(w_k - 2)$$

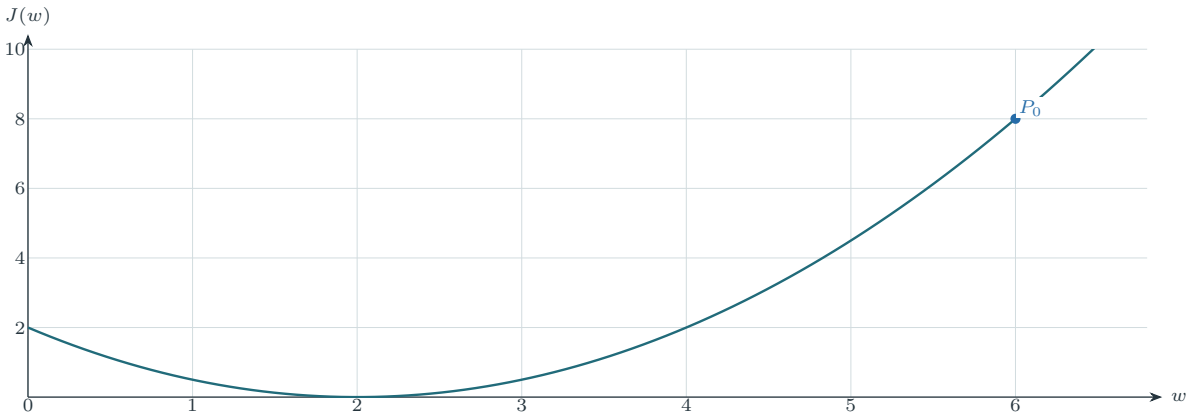
* For B, round each new w to two decimal places before the next row Table and plot values are approximate

k	Current w_k	Slope $w_k - 2$	Subtract $\alpha(w_k - 2)$	Next w_{k+1}
0	6			
1				
2				
3				
4				

8 Calculate each cost, plot $P_k = (w_k, J(w_k))$, and draw arrows in order $P_0 \to P_1 \to \cdots \to P_5$

* $J(w_k) = \frac{1}{2}(w_k - 2)^2$ $P_0 = (6, 8)$

$J(w_1)$	$J(w_2)$	$J(w_3)$	$J(w_4)$	$J(w_5)$



9 Check the **position and cost** after five updates

$w_5 \approx$ _____ $|w_5 - 2| \approx$ _____

$J(w_5) = \frac{1}{2}(w_5 - 2)^2 \approx$ _____ $J(w_0) = 8$

☐ At the minimum ☐ Stuck in another valley ☐ Still moving slowly

04 Learning Rate C

$J(w) = \frac{1}{2}(w-2)^2$ $J'(w) = w - 2$ $w_0 = 6$ $\boxed{\alpha = 0.5}$

10 Use the same rule to make **five updates**

$$w_{k+1} = w_k - 0.5(w_k - 2)$$

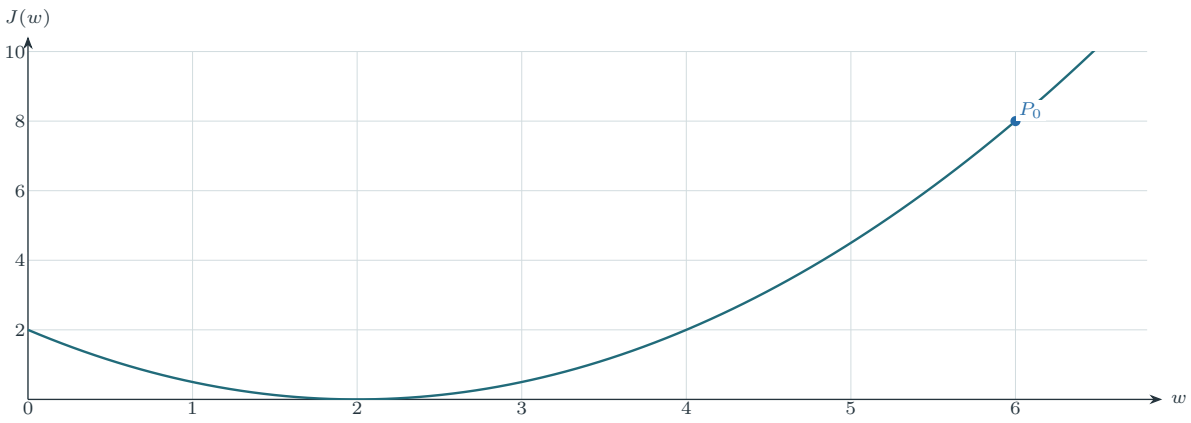
* Use the last value in a row as the current w in the next row

k	Current w_k	Slope $w_k - 2$	Subtract $\alpha(w_k - 2)$	Next w_{k+1}
0	6			
1				
2				
3				
4				

11 Calculate each cost, plot $P_k = (w_k, J(w_k))$, and draw arrows in order $P_0 \to P_1 \to \cdots \to P_5$

* $J(w_k) = \frac{1}{2}(w_k - 2)^2$ $P_0 = (6, 8)$

$J(w_1)$	$J(w_2)$	$J(w_3)$	$J(w_4)$	$J(w_5)$



12 Check the **position and cost** after five updates

$w_5 =$ _____ $|w_5 - 2| =$ _____ $J(w_5) = \frac{1}{2}(w_5 - 2)^2 =$ _____ $J(w_0) = 8$

☐ Closer ☐ Farther away ☐ Almost no movement

Explain why the steps get smaller even though the learning rate is fixed

13 Which case is closest to the minimum after five updates? _____

Does a small step alone establish convergence? _____